

Process & Decision Documentation

A3: Final Game

GBDA 302

April 11, 2026

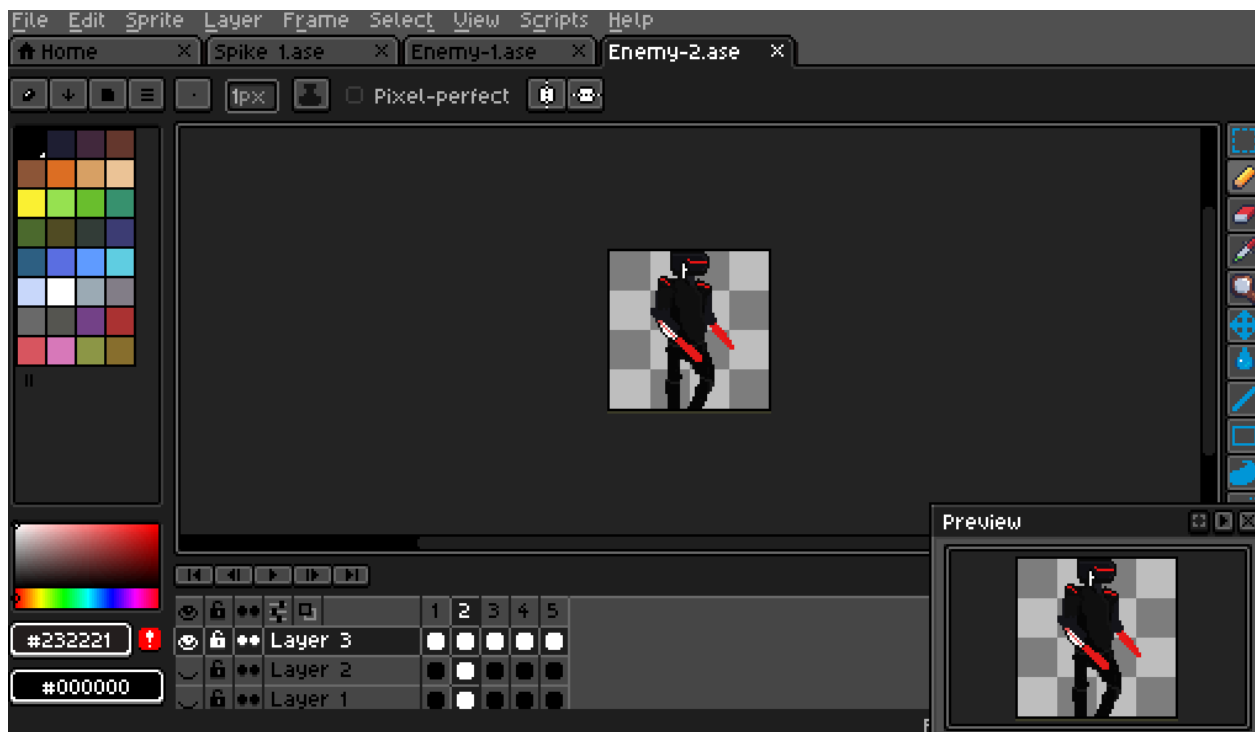
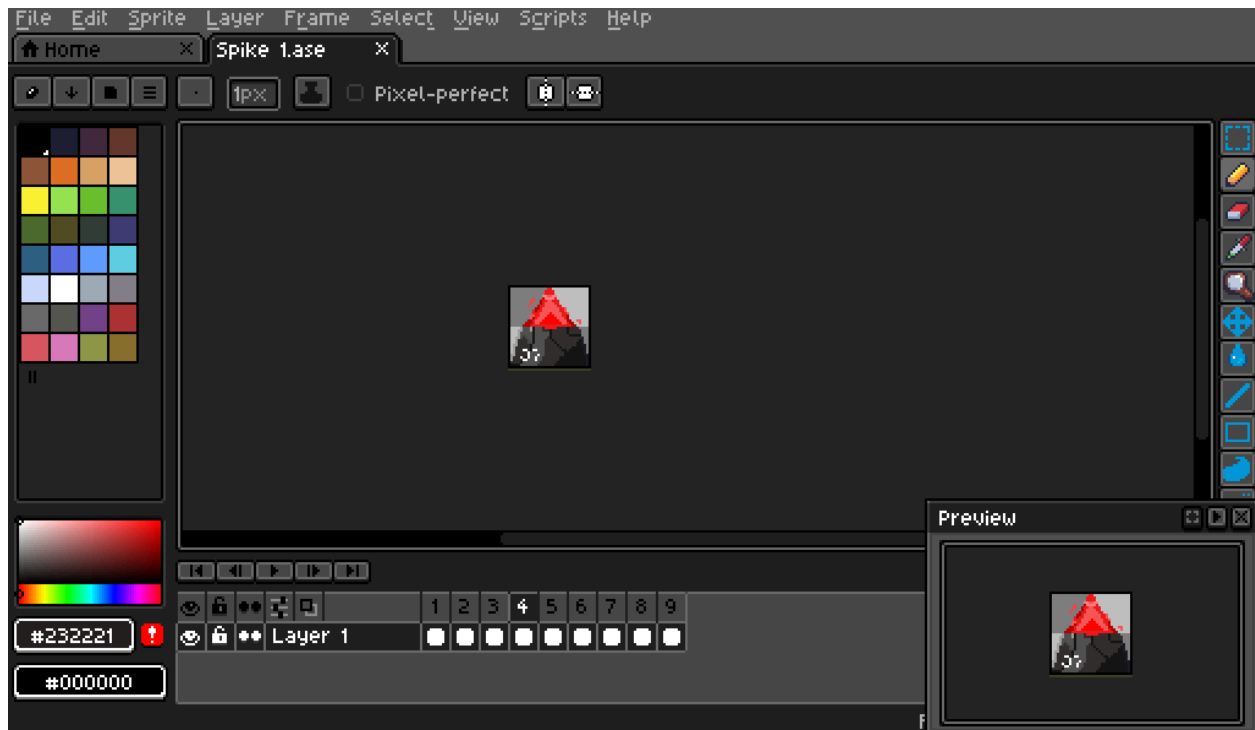
Emerson Sanchez, Jason Pham, Cohen Peters, Cameron Emsley-Drummond

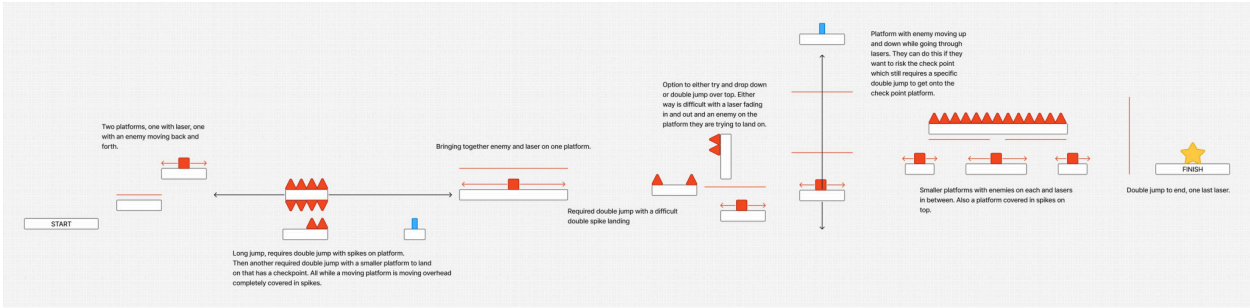
Group 9A

Process Overview Visualization

Below is a description of our group's design process for A3.

Asset Design





Phase 1:

Post Playtesting Adjustments (March 12, 2026)

- After a successful playtesting session, we sat down as a group and discussed what feedback we want to move forward with, and what to discard. We agreed we needed to make the game less difficult. We also wanted to add more levels to the game
- It was decided that the level we had for playtested would be adjusted by Jason to decrease level of difficulty

Phase 2:

Development (March 12 – April 1, 2026)

- Jason continued all p5.js development as Main Dev.
- Cameron and Cohen (Level Designers) began ideating and creating figma mockups of levels 2 and 3
- Emerson managed project progress, as well as created and designed animated sprites for character designs, power ups, enemies, and background
- All new features are developed on separate branches and reviewed before merging to main.
- Jason committed the following features by April 1: Updated enemy sprites, Levels 1+2, Level 3 + Assets, Wall Fix, Checkpoint Sprites + Levels, Level System

Phase 3:

Internal Playtesting (April 1, 2026)

- Team, lead by Cohen, tested the new levels, discussed what worked, and what we still want to work on for the final build
- Known bugs were fixed by Jason, and the game brought to near finished quality barring some additional details
- Build was playable from start to finish, we felt confident going into group playtesting.

Phase 4:

Playtesting (April 2, 2026)

- All members attended the in-class playtesting session (mandatory attendance).
- One member introduced the game concept to the group of playtesters
- Feedback was received from TAs, Professor, and peers

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- Further difficulty adjustments were suggested, positive feedback on visuals and controls of character

Phase 5:

Post-Playtesting Fixes (April 2 – April 7, 2026)

- Decided on what playtesting feedback to act on; everyone agreed on what we need ready for the final game, better difficulty tuning mainly
- Jason committed Playtesting Fixes on April 2, implementing feedback from the playtesting session from that day
- Tested fixes, ensuring no new bugs, game felt polished

Phase 6:

Final Build & Documentation (April 8, 2026)

- Full final game completed.
 - Process documentation created & finalized.
 - The GitHub Pages link is working and live.
 - Cohen (AI Verification) confirmed all GenAI usage was properly documented.
 - Jason committed Final tweaks on April 8, refining all levels, and adjusting mechanics/balancing
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Phase 7:

Polish, Showcase & Submission (April 12, 2026)

- Minor bug fixes and last touches finished.
- GitHub repo URL and GitHub Pages link submitted on LEARN by Jason before 11:59 PM.

Project/Assignment Decisions

Our group made several key decisions throughout the development of A3 that shaped the final product.

Decision 1 – Feedback After Midterm Playtest

What & Why	We knew what we wanted to create but now we finally had feedback from others outside the group. We received lots of positive feedback on our visual aesthetic and design, we also received constructive feedback on our difficulty balancing.
Who	Everyone in group had a say in this matter as we decided how much we should adjust aspects of our project
Effect it had in our game	We got lots of feedback on difficulty so that was a major decision we committed to here. We knew we needed to tone it down, and allow player skills to gradually get better instead of throwing them in the deep end. We also decided to go further with our visual aesthetic, by designing the enemies to feel like they belonged in this cyberpunk world.

Decision 2. – Adding More Levels

What & Why	With the gameplay mechanics down, and visual aesthetic further developed we knew we wanted to do more. As part of making the game more forgiving, we have essentially split up our game into 3 levels length wise.
Who	The entire team came to this conclusion after playtesting, Cameron and Cohen lead level design, Emerson designer additional sprites, and Jason developed it
Effect it had in our game	This decision dramatically changed our game for the better, creating a fuller experience that doesn't test the players' endurance as much.

Decision 3. – Difficulty Tuning

What & Why	We got strong feedback on many other elements of our game however, we failed to consider player skills and better difficulty tuning for our game. We knew we needed to make it a little more accessible so as not to turn people off of it or make them frustrated.
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Who	Essentially the whole team had to work on aspects of the game that led to it being a little less difficult. Cameron and Cohen tuned the levels to be more forgiving, Jason implemented changes within the code, and Emerson also had input on enemy design and placement.
Effect it had in our game	In our final game now we feel it is much more approachable and will not lead to players rage quitting or feeling frustrated. We tuned level design, platform locations, and enemy design to create an altogether stronger experience.

Role-Based Process Evidence

Entry 1	
Name	Emerson Sanchez
Role(s)	Project Lead, Animator (Shadow), QA Tester (Shadow)
Primary responsibility	Project coordination, milestone tracking, and designing/animating sprites
Goal of Work Session	Project coordination; kept the team on track against the milestone schedule from the updated A3 charter, facilitating decisions when there was disagreement, and ensuring every member knew their next steps at each phase of A3. Designed enemy sprites and background art for the game to further immerse the player in the cyberpunk world we built
Tools, Resources, or Inputs Used	• Instagram group chat for coordination • Group Charter milestone schedule • Teammates' progress updates • Group Charter milestone schedule and shared task documents • Libresprite for creating sprites

Entry 1 – (GenAI Documentation)

Name **Emerson Sanchez**

No GENAI usage

Summary of Process
(Human + Tool)

As a visual artist, I designed sprites that fit our game aesthetic using Libresprite. I animated and created the assets and designs using my previous knowledge and previously created artworks in the Sci/fi cyberpunk setting.

Decision Points &
Trade-offs

While I loved designing the enemy sprites, I had to ensure that they fit the tone of the game, as well as being able to reasonably fit on a platform in game. I had to focus my creativity on creating something that would exist in a preset world that we already built.

Verification & Judgement

I verified my designs by sending work-in-progress sprites to the team group chat, asking for feedback and shaping the current state of what the enemy designs look like as a team.

Limitations, Dead Ends, or
Open Questions

It certainly took some time coming up with enemy designs that would make it clear to the player that they are indeed enemies, but I also found the progress to be very creatively fulfilling. One thing I wanted to do was add more to the enemy animations; however, due to time constraints, I had to do what I could with the time I had.

Entry 2

Name	Jason Pham
Role(s)	Main Dev, Designer (Shadow)
Primary responsibility	All p5.js development, GitHub repository ownership, branch management, and final submission
Goal of Work Session	After our midterm submission, I felt really good going into A3. We knew what worked about our game, which was validating to hear, but I also knew the work ahead would be a solid amount, especially with adding a level system, as well as adding 2 more levels.
Role-Based Process Evidence	● Prompt engineered the core platformer, low vision effect, and focus mechanic phase by phase ● Created the level system ● Brought in sprites designed by Emerson including Enemy Sprites, Spike Sprites, and Checkpoint & Power up sprites ● Prompted and refined the Visual + Increased Difficulty pass (moving platforms, enemies, wall-jump, variable jump height) ● Prompted and refined Final Iteration before showcase (enemy stacking, updated focus mechanic), done by group online ● Committed some playtesting fixes after April 2 session. ● Finished with a cleanup on level 1 with adjustments to the echolocation mechanic, checkpoint visuals, and removing the double jump call. Also committed changes to level 3 to improve playability
Tools, Resources, or Inputs Used	● p5.js ● Visual Studio Code ● GitHub / GitHub Pages ● Web browser for testing and debugging ● Cohen's design specs and assets ● GenAI (coding support)

Entry 2 – (GenAI Documentation)

Name	Jason Pham
Date Used	During development of A3 (March 12, April 8, 2026)
Tool Disclosure	Claude Sonnet 4.6 / Opus 4.6 (Anthropic)
Purpose of Use	Used GenAI to assist with: • Post-playtesting difficulty rebalancing across Levels 1–3 • Implementing the level system (Level 1, 2, 3) with main menu / next-level flow • Adding moving platforms, enemy sprites, spike assets, and checkpoint sprites • Fixing the echolocation/focus mechanic to properly highlight platforms and enemies • Playtesting fixes (dark wall grace period on respawn, vision pickup reset, checkpoint placement) • Final cleanup round (double-jump bug, stationary echolocation, moving platform binding)
Summary of Interaction	• Level System 1+2 (Mar 31 – Apr 1) – prompted the three-level structure; Level 1 teaches core mechanics, Level 2 introduces enemies, Level 3 combines everything. Added next-level / main-menu UI button. Fixed focus highlighting. • Level 3 + Assets (Apr 1) – built out Level 3 and wired in Emerson's new spike sprites. Level design and bug fix pass. • Enemy & Checkpoint Sprites (Apr 1) – replaced placeholder graphics with Emerson's sprite sheets and hooked them into the existing classes. • Playtesting Fixes (Apr 2) – added respawn grace period for dark wall, reset vision pickup on death, moved Level 3 checkpoints off enemy platforms, added on-screen Level 1 instructions, enhanced checkpoint visual charm. • Cleanup Round 1 (Apr 8) – removed duplicate jump call, fixed echolocation wording, made echolocation work

	while stationary, bound moving platforms between stationary anchors. • 30%, 1.5s fade, 3.5s cooldown indicator • Visual + Increased Difficulty (Mar 4) - prompted, refined, built, and bug fixed: moving platforms, enemies, wall-jump mechanic, and variable jump height. Visual update (new assets, character + background effects, enemy interactions, HUD) done by group in-person • Final Iteration before showcase (Mar 5) - prompted, refined, built, and bug fixed: enemy stacking (if two enemies stack, platform becomes easier) and updated focus mechanic (held action, increased vision range, usable only while standing still on a platform). Visual redesign of menu, Jamie running and jumping animations, and dash bar repositioning done by group online
Human Decision Point(s)	I modified generated code after every prompt to fit our intended mechanics and fix bugs when GenAI solutions missed the mark. Level design layouts for all three levels were produced by Cohen and Cameron in Figma without prompting — I implemented those designs manually. The choice to add a respawn grace period, reset the vision pickup, and move checkpoints off enemy platforms were all calls made by the group after the April 2 playtest, not GenAI suggestions. Replacing the difficulty curve to be more forgiving was a group decision based on playtester feedback, not a GenAI recommendation.
Integrity & Verification Note:	All generated code was tested in the browser and verified through repeated in-team playtesting before being merged. New features were developed on separate branches and reviewed before merging to main. Cohen, as AI Verification shadow, confirmed each build visually matched design intent before final submission.
Scope of GenAI Use:	GenAI did not design the levels, the narrative, the sprite art, the difficulty balance, or the playtesting feedback decisions. Level design (all three levels) was done by Cohen and Cameron in Figma. All sprites, animations, and visual assets were created independently by Emerson. Playtesting was led by Cohen. Difficulty tuning and final-build polish decisions were made by the group in person.

Limitations or Misfires:

GenAI could not judge whether a level felt *fair* — only in-person playtesting surfaced the dark wall respawn trap and the Level 3 checkpoint-on-enemy-platform issue. The moving platform binding bug required several iterations to fix correctly. Sprite integration also needed manual tuning after the initial merge to align animation frames with movement.

Summary of Process
(Human + Tool)

I used Claude (Sonnet 4.6 / Opus 4.6) to speed up the p5.js work for A3. My loop was pretty consistent: prompt, read the output, rewrite the parts that didn't fit our code, test in the browser, commit. Nothing got committed straight from a prompt.

Most of the A3 work broke down into a few chunks. Around March 31 I got the three-level system in, with the main menu and next-level flow. On April 1 I wired in Emerson's enemy, checkpoint, and spike sprites and built out Level 3. After the April 2 playtest I spent a few days on fixes — the dark wall grace period on respawn, resetting the vision pickup on death, moving the Level 3 checkpoints off the enemy platforms, adding tutorial text to Level 1. Then on April 8 I did a cleanup round: killed a duplicate jump call, made echolocation work while standing still, and fixed the moving platforms so they stayed on their paths. Level design wasn't prompted — Cohen and Cameron did the Figma mockups and I built those by hand.

Decision Points &
Trade-offs

The hardest call was difficulty. The build we brought to the April 1 playtest was too punishing and we knew it. The question was how much to soften things without gutting the whole point of the low-vision mechanic.

I decided to keep the core constraint tight but loosen everything around it. The dark wall gets a grace period after respawn now. The vision pickup resets when you die. Echolocation works while you're standing still, so you can scan before committing to a jump. All of that makes the game more forgiving without pretending it's easy.

Level 3 checkpoints were another one. The original Figma layout put them on platforms that enemies patrolled. Technically fine, but in practice you'd respawn, immediately get hit, respawn, get hit. I moved them. We lose a bit of the "everything is dangerous" feeling but nobody actually enjoyed that version. With the moving platforms I locked them between fixed anchor points instead of letting them drift freely — it's a bit less interesting but

	the collisions actually work.
Verification & Judgement	Everything got tested in the browser before it hit main. I worked on branches and merged after reviewing. After any GenAI-assisted change I played through the affected level start to finish, because the focus mechanic in particular breaks in ways you don't notice until you try to use it. Cohen ran a full team playtest on April 1 before class, and another verification pass on April 8 before we submitted. Cohen was also checking builds against the Figma designs as part of the AI Verification role, which caught a few visual mismatches I would have shipped. The commit history with passing Actions checks is the real record — nothing went up without being tried first.
Limitations, Dead Ends, or Open Questions	The thing GenAI couldn't help with was feel. It could write me a moving platform, but it couldn't tell me if a jump was actually reachable once the low-vision effect was on top of it. That kind of thing only shows up when someone plays it. The moving platform binding took a few tries. The first couple of outputs either let the platforms drift off their paths or snapped them back hard enough to fling the player. The version that actually works needed me to hand-tune the interpolation. Sprite integration was similar — the initial merge played frames at a fixed interval regardless of what the player was doing, so I had to go back in and tie the animation states to actual movement. One thing I didn't solve: the echolocation cooldown indicator is small and easy to miss, and in playtests people kept mashing the key wondering why it wasn't working. A clearer HUD element is the obvious fix but I ran out of runway.

Entry 3	
Name	Cameron Emsley-Drummond

Role(s)	Level Designer, Documentation (Shadow)
Primary responsibility	Level design, process documentation, narrative
Goal of Work Session	Last time I focused on creating the vibe/narrative of the game, now my role has shifted to a level designer, as well as working on the process documentation. I developed level 2, with Cohen developing level 3.
Tools, Resources, or Inputs Used	• A3 assignment brief and README requirements • Lecture notes on empathy-driven design and representation • Process Documentation requirements • Group discussions and course concepts around disability representation

Entry 3 – (GenAI Documentation)

Name	Cameron Emsley-Drummond
<i>No GenAI was used for my role. All written content, in-game text, and process documentation were produced independently using course concepts, group discussions, and the A2 assignment brief.</i>	
Summary of Process (Human + Tool)	For A3 I developed levels 2 and 3 with Cohen in Figma. I focused my efforts on level design after working on the narrative in A2.
Decision Points & Trade-offs	Difficulty balance was key heading into designing the level. We knew we needed to push the player, but not push them too hard into rage-quitting or not enjoying the game. I think difficulty is such a hard line to balance, and I definitely had to decrease difficulty which potentially compromises our difficult game experience

Verification & Judgement	I verified choices by using my knowledge of 2D platformers played in class like celeste. Knowing how the devs working on that game balanced difficulty, and bracing players for larger spikes. I kept level design in games played in class front of mine while designing level 2 in figma
Limitations, Dead Ends, or Open Questions	A limitation was certainly that level 2 represents medium difficulty to us as a group, with level 3 being where we throw everything we can at the player. Level 2 represents a perfect balance, so that required myself to be very careful and take into deep consideration enemy placement, checkpoint placement, and length of the level so as to not frustrate or bore the player.

Entry 4	
Name	Cohen Peters
Role(s)	Level Designer, Playtester (Shadow)
Primary responsibility	Level Designing level 3
Goal of Work Session	My goal was creating level 3, as well as leading all playtesting activities. I playtested all the new features in the game to ensure we were following empathetic design principles and staying true to our assignment. I designed level 3, which is the ultimate test for our players, pushing them with all mechanics set up from previous levels.
Tools, Resources, or Inputs Used	• Figma • Celeste (design reference for visual disability representation in games) • Lecture notes on empathy-driven design and representation

Entry 4 – (GenAI Documentation)

Name	Cohen Peters
<i>No GenAI was used for my role.</i>	
Summary of Process (Human + Tool)	I was responsible for designing level 3 in figma, pushing the player to the limit in both a fun and challenging way to fully throw every game mechanic at them. I also took the lead on playtesting all features on the game.
Decision Points & Trade-offs	As a part of pushing the player to the limit in level 3, you need to have a steady hand at ensuring you aren't pushing them too hard, otherwise they will give up on the game and call it impossible. You need to make it just possible enough for them to get ahead by being patient, which meant I had to deal with trade-offs relating to how much difficulty I wanted to throw at the player.
Verification & Judgement	I verified much of my design decisions based on what we did with our mid term game, as well as Celeste which we played in class. Looking at our midterm game, it was called too hard, which was the inspiration for this final level. Celeste also was a great example of a challenging game, but still remaining rooted in empathy driven design.
Limitations, Dead Ends, or Open Questions	Designing in Figma was really great, but it likely would have been much easier being able to design with a pen attachment on an Ipad. I often see artists use that and envy the ability to just draw digitally, so I can share my work easily.

GenAI Documentation Summary (Group Level)

The following is a summary of all GenAI usage across the group for A3.

Team Charter Review - Claude Sonnet 4.6 (Jason Pham)

- **Date:** March 12, 2026
- **Tool:** Claude Sonnet 4.6 (Anthropic)
- **Purpose:** To review the A2 team charter and update it to reflect A3 roles, milestones, and GitHub workflow for the final game build.
- **Outcome:** Claude flagged role shifts for A3 (Cohen moving to Level Design / AI Verification, Cameron to Level Design, Emerson to Sprite Art & Project Management, Jason remaining Main Dev), GitHub branching workflow gaps, and milestone

checkpoints for the April 2 playtest and April 12 submission. The group reviewed the suggestions and adopted those that added value; repetitive or off-target ones were discarded. Final charter decisions were made by the group.

- **Prompt Author:** Jason Pham
- **Limitation:** The tool repeated content already in the A2 charter and suggested milestones that didn't fit our actual A3 timeline.

p5.js Code Development – Levels, Sprites, Mechanics (Jason Pham)

- **Dates:** March 31 – April 8, 2026
- **Tool:** Claude Sonnet 4.6 / Opus 4.6 (Anthropic)
- **Purpose:** To implement the three-level system, integrate new sprite assets, rebuild the focus/echolocation mechanic, and apply post-playtesting fixes.
- **Outcome:** Jason refined, bug-fixed, and tested each commit before merging. Key human revisions included the post-playtesting difficulty rebalance, moving Level 3 checkpoints off enemy platforms, and the final cleanup round (double-jump removal, stationary echolocation, moving platform binding). All level design was done by Cohen and Cameron in Figma and implemented manually.
- **Prompt Author:** Jason Pham
- **Transcript link:** See Appendix

Sections with No GenAI Usage:

- Level Design (Cohen Peters & Cameron Emsley-Drummond) – designed in Figma, no GenAI
- Sprite Art & Animation (Emerson Sanchez) – designed in Aseprite, no GenAI
- Playtesting Lead (Cohen Peters) – no GenAI
- Process & Decision Documentation – written independently by the group

Appendix

Full GenAI transcripts are included below for transparency and verification.

Transcript 1: Team Charter Review - Jason Pham

Prompt: "based on our current a2 team charter and assignment 3 is there anything we would need to update? we as a team are currently working well and responding to messages within a reasonable time. anything to improve to ensure we can effectively complete assignments on time?"

Summary: Claude identified role weight shifts for A3 (Cohen moving to Level Design / AI Verification, Cameron contributing level design alongside written content), GitHub workflow gaps around branching and review before merging to main, milestone checkpoint needs for the April 2 in-class playtest and April 12 showcase submission, and mandatory attendance re-acknowledgement for both dates. A follow-up prompt asked Claude to generate a revised charter reflecting these updates. Claude provided structural recommendations; all final content and decisions were made by the group.

What We Used or Changed: Adopted the milestone structure (April 2 playtest, April 12 submission) and the GitHub workflow framing around branching and review. Discarded

repetitive content that restated what was already in the A2 charter. Most final charter language was written independently by the group.

Transcript 3: p5.js Code Development - Jason Pham

Link: See GitHub commit history:

https://github.com/j8pham/A3_Group9A_FinalGame/commits/main/?fbclid=PAVERFWARHxNIIeHRuA2FlbQlXMAbZcnRjBmFwcF9pZA8xMjQwMjQ1NzQyODc0MTQAAacnLuvTNJSCbHMY9YGn9T0nxH7H_ahxz0VXSrbCboNkafFS36l4Y6tBNRPiiA_aem_kohYlMCKUjYXoQfJnVmdCQ

Prompts used (by phase):

- **Phase 1 – Level System (Mar 31):** "Introduce a level system with three levels in our p5.js platformer. Level 1 teaches core mechanics (movement, jump, echolocation) in a simple layout. Level 2 introduces enemies. Scaffold Level 3 as a placeholder. Add a UI button at level completion for 'Next Level' or 'Main Menu.' Fix the echolocation/focus feature so it properly highlights all platforms and enemies."
- **Phase 2 – Level 3 + Assets (Apr 1):** "Build out Level 3 combining enemies, moving platforms, and spikes. Import the new spike sprite assets and do a pass of bug fixes on Levels 1 and 2."
- **Phase 3 – Enemy & Checkpoint Sprites (Apr 1):** "Replace the placeholder enemy and checkpoint graphics with the new sprite sheets. Hook them into the existing enemy and checkpoint classes so animation frames cycle correctly."
- **Phase 4 – Playtesting Fixes (Apr 2):** "Based on playtest feedback: add a grace period on the dark wall after respawn, reset the vision/echolocation pickup on death, fix the Level 3 moving-platform overlap with static platforms, move Level 3 checkpoints off enemy-patrolled platforms, add on-screen Level 1 tutorial text, and enhance the checkpoint visual feedback."
- **Phase 5 – Cleanup Round 1 (Apr 8):** "Cleanup pass: remove the duplicate jump call, fix the echolocation tooltip wording, polish checkpoint visuals, make echolocation work while fully stationary, fix platform/enemy collision edge cases, and bind moving platforms between their stationary anchors."
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Summary: Claude generated initial code output for each commit. All testing, bug-fixing, feature refinement, and final integration decisions were made by Jason. Key human changes: rebalanced difficulty after the April 1 playtest, relocated Level 3 checkpoints off enemy-patrolled platforms, added a respawn grace period to the dark wall, reset the vision pickup on death, bound moving platforms between fixed anchor points after initial outputs caused drift, hand-tuned sprite animations to movement states, and removed a duplicate jump call that caused a double-jump bug.